Perron defects under reciprocal edge twists

Cycle rigidity and minimum realization sizes

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Abstract

We compare the Perron root of a reciprocal edge twist of a symmetric nonnegative matrix with an optimized lower bound at the untwisted Perron vector. A balanced gauge makes the left and right residuals equal, giving a Schur-complement formula for their spectral defect. Every finite nonzero order of vanishing is a multiple of four. On unicyclic support the only nonzero orders are $4, 8, 12$; the zero-defect class at nonzero circulation is exactly a pure cycle with constant diagonal and two-periodic edge weights. Every order $4m$ is realizable in dimension $2m + 2$, which is sharp in a specified two-cell class. The first three unrestricted minimum sizes are $3, 3, 4$; for zero-diagonal bipartite realizations the first two are $4, 5$.

1 Setup and relation to prior work

Let $B = B^\top \geq 0$ entrywise be $n \times n$, $n \geq 2$, with connected simple off-diagonal support $G = (V, E)$. Put $\lambda_0 = \rho(B)$ and normalize the positive Perron vector u by $u^\top u = 1$. For $\{i, j\} \in E$ let $c_{ij} = B_{ij} u_i u_j$. A real field h on oriented edges is antisymmetric. The twist $A(h)$ keeps the diagonal of B and has off-diagonal entries $B_{ij} e^{h_{ij}}$ on the support. Use $(\nabla\phi)_{ij} = \phi_j - \phi_i$ and set

$$F_h(\phi) = 2 \sum_{e \in E} c_e \{\cosh(h_e + \nabla_e \phi) - 1\}, \qquad E(h) = \min_{\phi_1 = 0} F_h(\phi). \tag{1}$$

Each edge sum counts an undirected edge once; the choice of orientation does not affect it. Neighbor sums are directed. Define $\Delta(t) = \rho(A(th)) - \lambda_0 - E(th)$.

The starting variational bound is a fixed-vector specialization of the spectral-radius formula of Altenberg [1]. The divergence-free sinh balance is related to the force/current decomposition of Kaiser, Jack and Zimmer [4]. In Markov coordinates $\operatorname{diag}(u)^{-1} B \operatorname{diag}(u) - \lambda_0 I$, the stationary law is u_i^2; fixed-density flow optimization and comparison with driven rates fit the variational framework of Chetrite and Touchette [3, Appendix B]. Fourth-order perturbation uses the same reduced-resolvent machinery as current-cumulant expansions [2]. Related current-fluctuation bounds and localization of complex generator eigenvalues [6, 5] concern different comparisons. Our focus is the order, rigidity and realization size of the particular deficit Δ. We make no claim of a new general variational principle or of global priority.

2 Minimizing gauge and perturbation

For a minimizing field $z = h + \nabla\phi$, write

$$k_i = u_i^{-2} \sum_j c_{ij} (\cosh z_{ij} - 1).$$

Theorem 1 (T1)**.** *The pinned minimizer in* (1) *exists and is unique. It satisfies* $\sum_j c_{ij} \sinh z_{ij} = 0$ *at every vertex. One has* $\rho(A(h)) \geq \lambda_0 + E(h)$*, with equality exactly when all* k_i *coincide.*

Proof. The Hessian in direction v is $2\sum_E c_e \cosh(z_e)(\nabla_e v)^2$. Connectedness makes it positive on nonzero pinned directions. On a sublevel $F_h \leq K$, every edge satisfies $|h_e + \nabla_e \phi| \leq \operatorname{arcosh}(1 + K/(2c_e))$. Summing the resulting bounds for $|\nabla_e \phi|$ along fixed paths from vertex 1 bounds every ϕ_i. Thus the nonempty sublevel $F_h \leq F_h(0)$ is compact, and strict convexity gives uniqueness. The pinned gradient equations give the divergence condition at all but the pinned vertex; antisymmetry gives it there too.

Gauge changes are diagonal similarities: $A(h + \nabla\phi) = D^{-1}A(h)D$, $D = \operatorname{diag}(e^{\phi_i})$. Write a positive right Perron vector of $A(h)$ as $u_i e^{q_i}$, scaled so that $q_1 = 0$. Divide its equation in row i by that entry, multiply by u_i^2, and sum. Pairing the two orientations gives $\rho(A(h)) - \lambda_0 = F_h(q)$. At the minimizing gauge, cancellation of the sinh terms gives $(A(z)u)_i/u_i = \lambda_0 + k_i$, and $\sum_i u_i^2 k_i = E(h)$. If the increments coincide, positivity identifies the displayed eigenvalue as the Perron root. Conversely equality makes q the minimizing gauge, so the Perron equation forces the increments to coincide. □

Let $\alpha = h + \nabla\psi$ be the harmonic field, characterized by $\sum_j c_{ij}\alpha_{ij} = 0$ with $\psi_1 = 0$; it is the unique minimizer of the positive grounded quadratic energy. For any potential f, $\sum_E c_e \alpha_e \nabla_e f = 0$. Let R be the inverse of $\lambda_0 I - B$ on $u^\perp$, extended by zero on u. Symmetry and simplicity of the largest eigenvalue give strict positivity of R on $u^\perp$, including in the bipartite case.

Theorem 2 (T2)**.** *Put* $K_{2,ij} = B_{ij}\alpha_{ij}^2/2$ *off the diagonal and* $K_{2,ii} = 0$*. For* $a = \sum_E c_e \alpha_e^2$*,* $b = \sum_E c_e \alpha_e^4/12$*, and* $g = K_2 u - au$*,*

$$E(th) = at^2 + bt^4 + O(t^6),$$
$$\rho(A(th)) = \lambda_0 + at^2 + (b + g^\top R g)t^4 + O(t^6).$$

The fourth-order coefficient of Δ *is nonnegative, vanishing exactly when* $K_2 u = au$*. The quantities are invariant under changing* h *by a gradient.*

Proof. Use the harmonic gauge. The divergence equations for the pinned minimizer $\eta(t)$ at field $t\alpha$ are analytic and their linearization in η at zero is the negative grounded weighted Laplacian. The analytic implicit-function theorem and uniqueness give an analytic odd solution, with $\eta'(0) = 0$ by harmonicity. In the energy expansion the fourth-order mixed term $2\sum_E c_e \alpha_e \nabla_e \eta_3$ vanishes. This yields a, b.

Write $A(t\alpha) = B + \sum_{\ell \geq 1} t^\ell K_\ell$, where $K_{\ell,ij} = B_{ij}\alpha_{ij}^\ell/\ell!$ off the diagonal. Odd K_ℓ are skew, even ones symmetric; $K_1 u = 0 = u^\top K_1$. The simple eigenpair is analytic near zero, with eigenvector normalized by $u^\top r(t) = 1$, positive near zero. Transposition makes the eigenvalue even. Comparing orders one and two gives $r_1 = 0$, $u^\top K_2 u = a$, and $r_2 = Rg$. Multiplying the fourth-order equation by $u^\top$ eliminates $K_1 r_3$ and all eigenvector normalization terms. It leaves $u^\top K_4 u + u^\top K_2 R g = b + g^\top R g$. Finally $u^\top g = 0$ and R is positive on $u^\perp$. □

3 Cycle current and the order mechanism

Theorem 3 (T3)**.** *On a unicyclic support with oriented cycle C and circulation $\Phi = \sum_C h_e$, the unique $J \in \mathbb{R}$ satisfying $\sum_C \operatorname{arsinh}(J/c_e) = \Phi$ gives*

$$E(h) = 2 \sum_{e \in C} (\sqrt{c_e^2 + J^2} - c_e).$$

At nonzero Φ, any vertex off the cycle forces a strict bound in T1. At $\Phi = 0$, the twist is diagonally similar to B.

Proof. The divergence-free current $c_{ij} \sinh z_{ij}$ vanishes on a bridge by summing divergences on one side. It is therefore constant in the cycle orientation. The sum of $\operatorname{arsinh}(J/c_e)$ is strictly increasing and onto. This choice of field has the right cycle sum and differs from h by a gradient, as seen by integrating on a spanning tree. Substitution gives the energy. At nonzero circulation the increments are positive on the cycle and zero outside, so they cannot all coincide. Zero circulation gives $J = 0$. □

Theorem 4 (T4)**.** *For the minimizing $z(t)$ at th set $e(t) = E(th)$ and $v_i(t) = u_i(k_i(t) - e(t))$. The functions e, k, v are even and real analytic near zero. If $v \equiv 0$ near zero then $\Delta \equiv 0$ there. Otherwise let $2m$ be the least degree of a nonzero Taylor coefficient v_{2m}. Then*

$$\Delta(t) = (v_{2m}^\top R v_{2m})\, t^{4m} + O(t^{4m+2}), \qquad v_{2m}^\top R v_{2m} > 0. \tag{2}$$

Proof. The minimizing field is analytic and odd by the implicit-function argument and gauge invariance. Put $M(t) = A(z(t))$. Balance cancels $\sinh z$ in both $M(t)u$ and $M(t)^\top u$. Thus relative to $\mathbb{R}u \oplus u^\perp$ the matrix has block form

$$M(t) = \begin{pmatrix} \lambda_0 + e(t) & v(t)^\top \\ v(t) & \mathcal{C}(t) \end{pmatrix}.$$

In particular $v \perp u$. The compression $\mathcal{C}(0)$ has no eigenvalue λ_0, so $\lambda(t)I - \mathcal{C}(t)$ is invertible near zero for $\lambda(t) = \rho(A(th))$. A positive right eigenvector has nonzero u-component; eliminating its perpendicular component yields

$$\Delta(t) = v(t)^\top (\lambda(t)I - \mathcal{C}(t))^{-1} v(t).$$

At zero the inverse is $R|_{u^\perp}$. Expanding gives (2), with the odd remainder coefficient absent by transposition symmetry. In particular $v_2 = g$; if it vanishes then $\Delta = O(t^8)$. □

4 Sharp rigidity on one cycle

Theorem 5 (T5)**.** *For unicyclic support and nonzero circulation, the defect has leading order $4, 8, 12$, with positive coefficient, or vanishes for every real t. The last case occurs exactly for a pure cycle with a constant diagonal and two-periodic edge weights. Odd cycles then have uniform weights. Any vertex outside the cycle forces order four, and all three finite orders occur.*

Proof. At an outside vertex the harmonic field vanishes on incident bridges, so $g_i = -a u_i \neq 0$. On a pure cycle write $r_i = 1/c_i$ for successive edge resistances. The current is an analytic local parameter since $J'(0) = \Phi / \sum_i r_i \neq 0$. The first three coefficients of $k_i(J)$ at degrees $2, 4, 6$ are

the incident sums of r, r^3, r^5 divided by u_i^2, multiplied by $1/2, -1/8, 1/16$, respectively. Changing from J to t preserves the first degree with a nonconstant coefficient.

If all three coefficients are constant across vertices, then u_i^2 is proportional to $r_{i-1} + r_i$. Ratios of the higher incident sums to the first yield common values on every adjacent pair (x, y):

$$Q = x^2 - xy + y^2, \qquad H = (x^5 + y^5)/(x + y) = Q^2 + xyQ - (xy)^2.$$

Compare (x, y) and (y, z). Equality of Q implies $(x - z)(x + z - y) = 0$. If $x \neq z$ then $y = x + z$; the difference of the H values is $y(x-z)(Q-y^2) = -xyz(x-z) \neq 0$, impossible. Thus $r_{i+1} = r_{i-1}$, and the first test makes u uniform. The Perron equation gives constant diagonal and two-periodic weights of B. Conversely those weights and a constant diagonal yield uniform u and identical increments at every J; T1 then gives global equality. If any of the three tests fails, the first such test and T4 give the stated order. The following exact witnesses prove sharpness. □

Exact witnesses. Use the unnormalized convention $U > 0$, $M = U^\top U$, and positive conductances C_i around a pure cycle. Define $B_{ij} = C_{ij}/(U_i U_j)$ on edges and $B_{ii} = \lambda_0 - \sum_j C_{ij}/U_i^2$. In each row below the diagonal is nonnegative and $BU = \lambda_0 U$. Write $r_i = 1/C_i$, $S = \sum_i r_i$; set $h_i = r_i/S$ in the cycle orientation. The conductances in (1) are C_i/M.

Order	U	C around the cycle	λ_0
4	$(1, 1, 1)$	$(1, 1, 2)$	5
8	$(2, 3, 3)$	$(1/2, 1/7, 1/2)$	1
12	$(3, 3, 4, 4)$	$(3/2, 3/25, 3/23, 3/25)$	1

For the first nonconstant incident moment $F_i = (r_{i-1}^{2j-1} + r_i^{2j-1})/U_i^2$, the leading $\sqrt{M}\, v_{2j}$ is $b_j U_i (F_i - \bar{F})/S^{2j}$ in row i, where $\bar{F} = \sum_i U_i^2 F_i/M$ and $b_1 = 1/2, b_2 = -1/8, b_3 = 1/16$. Indeed the lower moments are constant and cancel after centering, and the current for C has first coefficient $1/S$. In the eighth-order example $S = 11$, the first moment is constant and the third is $(4, 39, 39)$. In the twelfth-order example $S = 25$, the first and third are constant, 1 and $193/3$; the fifth takes two values whose difference is $8050/27$. Substitution gives these certificates (each $x \perp U$):

$4j$	x	$\sqrt{M}\, v_{2j}$	Rx
4	$(-1, 2, -1)$	$x/75$	$x/3$
8	$(3, -1, -1)$	$105x/(4 \cdot 11^5)$	$36x/11$
12	$(4, 4, -3, -3)$	$161x/(18 \cdot 25^6)$	$48x$

The coefficients from (2) are respectively $2/16875$, $99225/(8 \cdot 11^{11})$, $103684/(27 \cdot 25^{12})$. The resolvent relations are verified by multiplying by $\lambda_0 I - B$; the factor $1/M$ in the quadratic form converts the unnormalized vectors to unit-Perron coordinates.

5 Every multiple of four and smaller realizations

Theorem 6 (T6–T7)**.** *For every integer $m \geq 1$, rational matrices with positive diagonal and integer twists realize the coefficient $1/[2((2m)!)^2]$ at order $4m$, in dimensions $4m + 2$ and $2m + 2$. Each construction has a connected zero-diagonal bipartite lift preserving the entire defect, in dimensions $8m + 4$ and $4m + 4$, respectively. The dimension $2m + 2$ is minimal within the two-cell class specified below.*

Proof. For $1 \le a \le m$ put

$$d_a = \frac{1}{a^2 \prod_{b \neq a}(a^2 - b^2)}, \qquad \sum_{a=1}^{m} d_a a^{2j} = \begin{cases} 0, & 1 \le j < m, \\ 1, & j = m. \end{cases}$$

The identity is the highest-coefficient comparison for Lagrange interpolation of x^{j-1} at the distinct nodes a^2. Let $D(t) = \sum_a d_a(\cosh(at) - 1) = t^{2m}/(2m)! + O(t^{2m+2})$.

For the first construction take two cyclic cells of size $2m + 1$. The oriented edges $i \to i + a$ for all vertices i and $1 \le a \le m$ cover each unordered pair exactly once. Use twist a and weights $w_a^{\pm} = 1 + \max(\pm d_a, 0)$ in the two cells. Connect corresponding vertices by weight-one edges of zero twist. Choose a common row sum $\lambda_0 = 2 + 2\sum_a \max(w_a^+, w_a^-)$ by filling the diagonal. All diagonal entries are at least one. The Perron vector is uniform. At zero potential, each color balances in the sinh equation at every vertex. The row increments $k_{\pm} = 2\sum_a w_a^{\pm}(\cosh(at) - 1)$ are constant on the cells and $E = (k_+ + k_-)/2$. The cell-constant quotient has diagonal $\lambda_0 - 1 + k_{\pm}$ and off-diagonal entries one. Its positive Perron eigenvector lifts, so its Perron root is the full one. Therefore $\Delta(t) = \sqrt{1 + D(t)^2} - 1$, proving the claimed order and constant.

For the smaller construction put the positive and negative indices of d_a in separate cells, with k_+, k_- indices respectively and $L_{\pm} = 2k_{\pm} + 1$ vertices in each cell. The r-th index a in a cell is assigned to all cyclic edges $i \to i + r$, with weight $|d_a|$ and twist a. A cell without indices is a singleton. Write $n = L_+ + L_-$, $p = L_+/n$, $q = L_-/n$, and give every cross-cell edge weight $w = 8L_+L_-/n^3$ and twist zero. Then $\gamma = nw = 8pq$, with cross-cell row sums $q\gamma, p\gamma$. Choose λ_0 one larger than the maximum off-diagonal row sum and fill the diagonal. Again the Perron vector is uniform, zero potential minimizes the energy, and $E = pk_+ + qk_-$. Now $d = k_+ - k_- = 2D$. The quotient after subtracting $(\lambda_0 + E)I$ is

$$\begin{pmatrix} -q\gamma + qd & q\gamma \\ p\gamma & -p\gamma - pd \end{pmatrix}.$$

Its trace is $-\gamma + (q - p)d$ and determinant $-pqd^2$. The nonnegative root Δ satisfies $\Delta^2 + [\gamma + (p - q)d]\Delta - pqd^2 = 0$. Expansion of the larger root near zero gives $\Delta = (pq/\gamma)d^2 + O(d^3)$, and the moment identity yields the same coefficient $1/[2((2m)!)^2]$ and remainder $O(t^{4m+2})$.

For either construction use the bipartite lift $\widehat{B} = \left(\begin{smallmatrix} 0 & B \\ B & 0 \end{smallmatrix}\right)$ with off-diagonal blocks $A(th)$ in the twist. The edges from the positive diagonal connect corresponding vertices across sheets; the original support ensures connectedness. A positive Perron vector $r(t)$ lifts to $(r(t), r(t))$, so the root is unchanged. The normalized base vector is $(u, u)/\sqrt{2}$. The energy and its minimizer modulo constants are invariant under swapping sheets; if the potentials differed by a constant under this involution, two swaps would double it, so it is zero. At the symmetric potential, two copies of each old edge contribute half its old energy each, and the diagonal edges contribute zero. Hence the defect is identical.

For the lower bound consider only realizations with uniform u, two nonempty cells, zero twists across cells, zero minimizing potential near 0, and increments constant in each cell near 0. The analytic functions $\cosh(at) - 1$ for distinct $a > 0$ are linearly independent by their first even coefficients (a Vandermonde matrix). Thus every vertex in a cell sees all $s_{\pm}$ nonzero twist magnitudes of that cell. Independence of the corresponding $\sinh(at)$ makes each magnitude balance separately, requiring at least two incident edges per magnitude at each vertex. Each cell has at least $2s_{\pm} + 1$ vertices. The difference of increments has at most $s_+ + s_-$ magnitudes. To begin at degree $2m$ it needs at least m magnitudes, by the same Vandermonde argument. T4 identifies this degree when the defect has order $4m$; thus $n \ge 2m + 2$. □

6 Minimum dimensions and constant-diagonal rigidity

Theorem 7 (T8)**.** *Let N_m be the least dimension allowing a strictly positive leading term of order $4m$ under the initial assumptions. Then $N_1 = 3,\ N_2 = 3,\ N_3 = 4$. For $m \geq 4$, $4 \leq N_m \leq 2m+2$.*

Proof. On a tree every field is a gradient, so dimension two is impossible. A connected three-vertex support is a tree or a triangle. On a triangle with nonzero circulation, if the first two tests from T5 are constant, $Q(x, y)$ has the same value on each pair of its three resistances. For x the smallest, $Q(x, s)$ is strictly increasing on $s \geq x$; hence the other two resistances are equal, say y. Then $Q(x, y) = Q(y, y)$ gives $x(x - y) = 0$, forcing all equal. The first test gives uniform u, and T5 gives equality rather than order twelve or higher. The three witnesses in Section 4 attain the lower bounds, and T7 supplies the general upper bound. □

Theorem 8 (T9)**.** *On unicyclic support with constant diagonal, the only possibilities are order four or identically zero, with the same nonzero-circulation equality class as T5. If Z_m is the minimum for zero-diagonal bipartite realizations of order $4m$, then*

$$Z_1 = 4, \qquad Z_2 = 5, \qquad 5 \leq Z_3 \leq 8.$$

A five-vertex $K_{2,3}$ witness has coefficient $5/512$ at order eight.

Proof. A common diagonal can be subtracted without changing u, c, E or the defect: both positive Perron eigenvalues decrease by exactly that diagonal. Zero circulation and outside vertices are covered by T3 and T5. On a zero-diagonal pure cycle with nonzero circulation let $J_0 = \Phi / \sum_E c_e^{-1}$. If $g = 0$, the harmonic field $\alpha_e = J_0/c_e$ gives

$$J_0^2(c_{i-1}^{-1} + c_i^{-1}) = 2au_i^2, \qquad \lambda_0 u_i^2 = c_{i-1} + c_i.$$

Cancel the positive sum to get a constant product $c_{i-1}c_i = \lambda_0 J_0^2/(2a)$. Hence the conductances are two-periodic, u is uniform and T5 gives equality. Otherwise T2 gives order four.

For the witnesses, use positive rational C_{ij} on edges, masses $\mu_i = \sum_j C_{ij}$, and $U_i = \sqrt{\mu_i}$. Set $B_{ij} = C_{ij}/(U_i U_j)$ on edges, with zero diagonal, so $BU = U$; $M = \sum_i \mu_i$ and $u = U/\sqrt{M}$. For a four-cycle use $C = (1, 1, 1, 2)$, with harmonic $h = (2/7, 2/7, 2/7, 1/7)$. The masses are $(3, 2, 2, 3)$. The centered second-order increment is $f_2 = (-2, 3, 3, -2)/245$. In these coordinates $\sum_j C_{ij} f_{2,j}/\mu_i = f_{2,i}/6$ and $\sum_i \mu_i f_{2,i}^2 = 60/245^2$. The fourth-order coefficient is $(6/5)(60/245^2)/10 = 36/300125$.

For $K_{2,3}$, orient its three two-edge paths from L to R through M_1, M_2, M_3. Give the six consecutive edges conductances $(1, 4, 4, 1, 4, 4)$ and twists $(2, 1/2, 1/2, 2, -1, -1)$. Path currents are $2, 2, -4$, so the twist is harmonic. The masses are $(9, 9, 5, 5, 8)$, and $k_2 = 1/2$ at every vertex. The cubic potential coefficient η_3 solves $L_C \eta_3 = \operatorname{div}(Ch^3)/6$, where L_C is the positive weighted graph Laplacian and divergence sums outward fields. Up to a constant it is $(5/24, -5/24, -3/8, 3/8, 0)$: substitution on the three paths verifies the equation. The fourth-order increments follow by summing $C_e[h_e \nabla_e \eta_3 + h_e^4/24]/\mu_i$ at each vertex. They are $(3/32, 3/32, -1/32, -1/32, 1/4)$, of weighted mean $3/32$. Thus $f_4 = (0, 0, -1/8, -1/8, 5/32)$. Each $\sum_j C_{ij} f_{4,j} = 0$; consequently $B(Uf_4) = 0$ and $R(Uf_4) = Uf_4$. The defect coefficient is $\sum_i \mu_i f_{4,i}^2/36 = 5/512$.

Connected bipartite supports on at most three vertices are trees, and on four vertices the only non-tree possibility is a four-cycle. The dichotomy excludes order eight or higher on that zero-diagonal cycle. The witnesses attain $Z_1 = 4$ and $Z_2 = 5$, and lifting the order-twelve four-cycle gives the upper bound $Z_3 \leq 8$. □

7 Scope and reproducibility

The proof covers the precise revision-5 statements T1–T9 of `C-e83ae1e1d8f8447c846f8208`. The general $2m + 2$ upper bound does not prove global optimality, Z_3 is not determined, and the bipartite minima do not classify unrestricted zero-diagonal realizations. All asymptotic remainders are for fixed B, h.

The companion proof files include further exact certificates, including a direct characteristic-polynomial check for a triangle with a leaf. Five Python reproduction programs and the separate bounded search give finite checks, not proofs of universal assertions. The exact clarified proof snapshot passed two independent full paper reviews and a separate bounded counterexample and boundary review (fixed record `F-85c183ad590bceac57b760cdf542e402`). The review and drafting process used AI assistance. This is a versioned paper-track validation, with no Lean formalization or claim of journal peer review. The article and original exposition are available under CC BY 4.0; the supplied reproduction scripts are MIT-licensed.

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